

Contact

Not sure what to put here.
saniasingh721@gmail.com

www.linkedin.com/in/saniasingh21
(LinkedIn)
[saniasingh.in](#) (Personal)

Top Skills

Time-Dependent Schrödinger Equation (TDSE)

Finite Element Methods

Dirac (Bra-Ket) Notation

Languages

English (Full Professional)

Italian (Elementary)

German (Elementary)

Hindi (Native or Bilingual)

Punjabi (Elementary)

Sanskrit (Elementary)

Certifications

Grade 4 Guitar (Merit)

Grade 6 Vocals (Distinction)

Honors-Awards

Undergraduate Scholarship

Rank 2 in Cohort

Top Scorer: Chemistry

Perfect Score: Science

Perfect Score: Social Science

Sania Singh

Mathematician. Physicist. Software Engineer | MSc Imperial College London | BSc Miranda House
London, England, United Kingdom

Summary

Sania Singh is an applied mathematician, physicist, software engineer, musician, producer, and writer. Her work is grounded in what she calls the MPC arc: Mathematics, Physics, and Computing. She builds models from first principles, combining mathematical rigour with practical engineering to understand and solve complex problems. She holds an MSc in Applied Mathematics from Imperial College London, specialising in Scientific Computing and Machine Learning. Her master's research focused on plasma simulation using finite element methods implemented in Firedrake and Python. She previously completed a BSc (Honours) in Physics at Miranda House, University of Delhi, where she studied topics including numerical solutions of the Schrödinger equation, particle physics, classical dynamics, and computational modelling. Her research interests lie at the intersection of scientific computing, artificial intelligence, computational physics, and healthcare systems. She is interested in developing mathematical models, numerical methods, and reliable software that translate theory into practical tools for research and engineering. Alongside her scientific work, Sania is a musician, songwriter, producer, and writer. She writes about science, creativity, identity, and the process of building a meaningful life through her publication, Productive Overthinking. After performing with Imperial College London's Live Music Society, she is now producing her debut album, written and produced entirely by herself. Mathematics. Music. Writing. Three disciplines. One body of work.

Experience

Unnamed

Lead Data Architect - AI Products & Systems

January 2026 - March 2026 (3 months)

Silicon Valley

Sania Singh built production AI software and data infrastructure for a Silicon Valley AI startup. Further details available upon request.

Department of Mathematics at Imperial College London

Computational Mathematics Researcher

January 2025 - September 2025 (9 months)

London Area, United Kingdom

Sania Singh conducted independent research in computational plasma physics under the supervision of Professor Colin Cotter in the Department of Mathematics at Imperial College London. She developed finite element solvers for the collisionless Vlasov-Poisson system using Firedrake, implementing discontinuous Galerkin discretisations, Hermite quadrature, and SSPRK3 time integration to investigate hybrid multistream formulations for kinetic plasma simulation. Her research combined finite element methods, numerical analysis, and computational partial differential equations, with convergence studies, moment-based error analysis, and numerical validation against full phase-space finite element models.

Miranda House, University Of Delhi

Computational QM Researcher

July 2022 - June 2023 (1 year)

New Delhi

Sania Singh developed and compared numerical methods for solving the time-independent Schrödinger equation across the hydrogen atom, screened (Yukawa) Coulomb potential, and one-dimensional harmonic oscillator (including an anharmonic perturbation). Using Python, SCILAB, and Wolfram Mathematica, she implemented finite-difference Hamiltonian diagonalisation, the Matrix Numerov Method, and the Rayleigh-Ritz variational method with Gram-Schmidt orthogonalisation, validating numerical solutions against analytical results and published reference values where available.

Education

Imperial College London

Master of Science - MS, Applied Mathematics (Scientific Computing and Machine Learning) · (September 2025)

Miranda House, Delhi

Bachelor of Science - BS, Physics · (June 2023)

Army Public School Noida
Class XII, Science (PCM+CS)

Army Public School Noida
Class X